

Data quality findings

1. JSON files did not follow a standardized structure

Problem: Some JSON files did not comply with a valid or consistent JSON format, making automated ingestion and parsing difficult.

Solution: Implement a preprocessing and validation stage to standardize the JSON structure before loading the data. Use parsing and validation tools to ensure schema consistency across all files.

2. Some fields were not normalized

Problem: Several columns contained inconsistent values, formats, or naming conventions, affecting data quality and analysis reliability.

Solution: Apply data normalization rules during the transformation process, including standardizing text formats, categorical values, date formats, and naming conventions.

3. Some columns contained unnecessary quotation marks

Problem: Certain fields included extra quotation marks, generating inconsistencies and affecting filtering, joins, and aggregations.

Solution: Perform data cleansing procedures to remove unnecessary special characters and quotation marks during the ETL process.

4. Some columns contained blank values instead of NULL values

Problem: Several records contained empty strings ("") rather than proper NULL values, causing inconsistencies in analysis and KPI calculations.

Solution: Replace blank or empty string values with standardized NULL values during data transformation to improve data integrity and analytical accuracy.

5. Some NULL values were stored as text

Problem: Some records contained the literal text "NULL" instead of actual database NULL values, affecting filtering and aggregation logic.

Solution: Identify and convert text-based NULL representations into proper NULL values during the cleansing and loading stages.

Conclusions for Q2 and Q3

Q2 Conclusion:

During Q2, the analysis showed that the total number of flights operated was lower than the estimated projections. This indicates a decrease in flight activity compared to expectations, which may be related to operational limitations, demand fluctuations, or scheduling adjustments. Monitoring these variations is important to improve forecasting accuracy and operational planning.

Q3 Conclusion:

In Q3, flight operations continued below the estimated levels, although the performance trend should continue to be monitored to identify possible recovery patterns or persistent operational challenges. The dashboard analysis provides visibility into flight status, delays, and operational behavior, supporting better decision-making and performance evaluation.